

# Ionospheric Electron Density Reconstruction Based on Space-borne SAR in Alaska Regions

Cheng Wang, Li-Ming Wang, Hai-Sheng Zhao, Peng Xiao, Liang Chen, and Le Cao

**Abstract**—The use of low-frequency spaceborne fully-polarized synthetic aperture radar (SAR), such as the ALOS PALSAR, to detect the structure of ionospheric has been widely validated in recent years. In terms of total electron content (TEC) retrieval, SAR exhibits significant advantages over GPS with regard to both accuracy and resolution at high-latitude. However, TEC cannot reflect the vertical structure of the ionosphere and still has limitations. On this basis, this letter presents a study into the application of fully-polarimetric PALSAR in computerized ionospheric tomography (CIT), which extends the horizontal ionospheric structure to the vertical, i.e., the three-dimensional (3-D) ionospheric electron density (IED) distribution. The reconstruction results from three groups of PALSAR measured data in Alaska are compared with the results from the nearby incoherent scatter radar (ISR). The accuracy has improved by 28.9%, 33.2%, and 47.6% respectively compared to the ionosphere model. This shows that the PALSAR has the ability to not only determine the horizontal structure of the ionosphere but also accurately reconstruct the 3-D IED, making it a promising approach for ionosphere sounding at high-latitude.

**Index Terms**—Ionospheric electron density, computerized ionospheric tomography, synthetic aperture radar.

## I. INTRODUCTION

THE ionosphere will cause significant effects on the low-frequency spaceborne synthetic aperture radar (SAR) signals in terms of time delay, scintillation, and Faraday rotation angle (FRA) [1]. Since the launch of the Phased Array L-band SAR on board the Advanced Land Observing Satellite (ALOS PALSAR), these issues have also received attention and different suppression strategies have been taken [2], [3], [4], [5]. However, the more obvious the ionospheric effect is, the more feasible it is to use the echo to retrieve the ionosphere. The SAR system's high resolution enable it to detect fine ionospheric features that are challenging to observe using

conventional approaches. Due to the combined effects of the ionosphere and the geomagnetic field, PALSAR echoes contain obvious FRA at high-latitude, which provides a new means for polar ionospheric research. Pi et al. [6] verified that the PALSAR can detect kilometer-scale FRA enhancement associated with auroral arcs. Wang et al. [7] verified that the accuracy of the total electron content (TEC) retrieved by the PALSAR is also significantly better than that of GPS in Alaska.

However, FRA or TEC still shows the horizontal information and cannot obtain the three-dimensional (3-D) ionospheric electron density (IED). Hu et al. [8] and Wang et al. [9] conducted theoretical and simulation studies on 2-D computerized ionospheric tomography (CIT) based on geosynchronous (GEO) SAR and PALSAR, but experimental validation and 3-D reconstruction are not carried out. Zhu et al. [10] implemented a unified correction to the International Reference Ionosphere (IRI) model with altitude using PALSAR, however, the approach remains a 1-D IED retrieval. Wang et al. [11] used the TEC values obtained from PALSAR to uniformly correct the empirical models at different altitudes, thereby enhancing the realism of the initial values in the iterative process and ultimately improving the accuracy of GPS-based CIT.

It can be seen that similar to GPS, the PALSAR can also retrieve the TEC values at different angles, which can therefore be used as a new data source for CIT. In addition, compared with GPS, the PALSAR signals propagate in two-way, and do not require ground receiving station, which will be easier for the implementation of CIT. On this basis, this letter does not consider GPS, and only uses PALSAR to reconstruct the IED distribution. Section II explains the fundamental principles. In Section III, three groups of PALSAR measurements from different times and locations in Alaska are taken to achieve IED reconstruction, and the results are compared with the results of the nearby Poker Flat incoherent scatter radar (ISR). The results confirmed that PALSAR is capable of 3-D reconstruction, and the obtained IED can reflect the true spatial distribution. Section IV discusses the relevant conclusions.

## II. METHODOLOGY

### A. FRA Retrieval based on PALSAR

When the signal passes through the ionosphere twice, the FRA will cause the polarization state of the SAR to rotate accordingly, affecting the polarization characteristic quality of the SAR data. Therefore, the key is to retrieve accurate path FRA or TEC value from the echo. The measured scattering matrix  $\mathbf{Z}$  can be written as [12]

Manuscript received 16 March, 2024. This work was supported by the National Natural Science Foundation of China (NSFC) under Grants 42074225, 62101433, and 62201529. (Corresponding author: Cheng Wang.)

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$$\begin{bmatrix} Z_{hh} & Z_{vh} \\ Z_{hv} & Z_{vv} \end{bmatrix} = \begin{bmatrix} \cos \Omega & \sin \Omega \\ -\sin \Omega & \cos \Omega \end{bmatrix} \cdot \begin{bmatrix} S_{hh} & S_{vh} \\ S_{hv} & S_{vv} \end{bmatrix} \cdot \begin{bmatrix} \cos \Omega & \sin \Omega \\ -\sin \Omega & \cos \Omega \end{bmatrix}, \quad (1)$$

where  $\mathbf{S}$  is the true scattering matrix. FRA and can be defined as

$$\Omega = 2.365 \times 10^4 H \cos \theta_H \text{TEC} / f^2, \quad (2)$$

where  $f$  is the frequency,  $H$  is the magnitude of the ambient magnetic field,  $\theta_H$  is the angle between the propagation vector and the magnetic field vector.  $\bar{\bullet}$  is the path's mean value.

Considering the residual errors after PALSAR calibration (The noise equivalent sigma zero is approximately -27 dB. The amplitude imbalance ranges between 0.5 and 1 dB, while the phase imbalance remains within 1°-5°. Published values for crosstalk range from -25 to -45 dB [3]), there are several FRA estimators based on the scattering matrix or covariance matrix [13], [14], [15], [16]. Among them, the algorithm based on the cross-polarization correlation term of circular polarization basis has the best robustness, which is the preferred algorithm in this letter. The expression of the algorithm can be written as

$$\begin{bmatrix} Z_{ll} & Z_{lr} \\ Z_{rl} & Z_{rr} \end{bmatrix} = \begin{bmatrix} 1 & j \\ j & 1 \end{bmatrix} \cdot \begin{bmatrix} Z_{hh} & Z_{vh} \\ Z_{hv} & Z_{vv} \end{bmatrix} \cdot \begin{bmatrix} 1 & j \\ j & 1 \end{bmatrix}, \quad (3)$$

where the cross-polarized terms are

$$\begin{cases} Z_{lr} = (S_{hh} + S_{vv}) \exp(j\pi/2 - j2\Omega) \\ Z_{rl} = (S_{hh} + S_{vv}) \exp(j\pi/2 + j2\Omega) \end{cases}. \quad (4)$$

Thus, the FRA can be obtained by

$$\Omega_B = -1/4 \cdot \arg(Z_{lr} \cdot Z_{rl}^*). \quad (5)$$

The slant TEC is further calculated according to (2).

### B. CIT based on the PALSAR

CIT is first proposed by Austen et al. [17], and subsequent advancements have been made using polar-orbiting satellite data, beacon data, and GNSS data [18], [19], [20]. In this letter, a new CIT data source, i.e., the PALSAR is used for CIT. As shown in Fig. 1, the ionosphere is divided into  $J$  voxels along the longitude, latitude, and altitude directions. The PALSAR performs ground imaging along the azimuth direction, with a total of  $M$  radar images (i.e.,  $M$  PALSAR rays). The signals are influenced by both the ionosphere and the geomagnetic field. The intersection points where each ray entries and exits the ionosphere are denoted by the black solid dots. According to (2) and (5), the TEC value along the path from the image center to satellite can be calculated, which is also the summation of discretized IED along the path, i.e.,

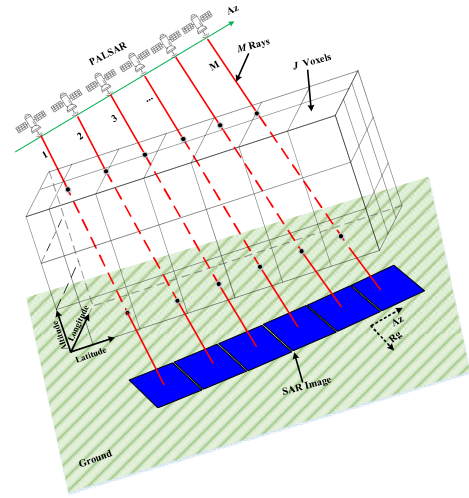
$$\text{TEC}_i = \sum_{j=1}^J a_{ij} x_j + \varepsilon_i, \quad (6)$$

where  $\text{TEC}_i$  is the TEC value retrieved by the  $i$ th PALSAR ray ( $i=1, \dots, M$ ),  $x_j$  is the IED of the  $j$ th voxel ( $j=1, \dots, J$ ) to be estimated. With each voxel,  $x_j$  can be assumed as constant.  $a_{ij}$  is the intercept of the  $i$ th ray in the  $j$ th voxel, and  $\varepsilon_i$  is the discrete and PALSAR residual error of the  $i$ th ray path.

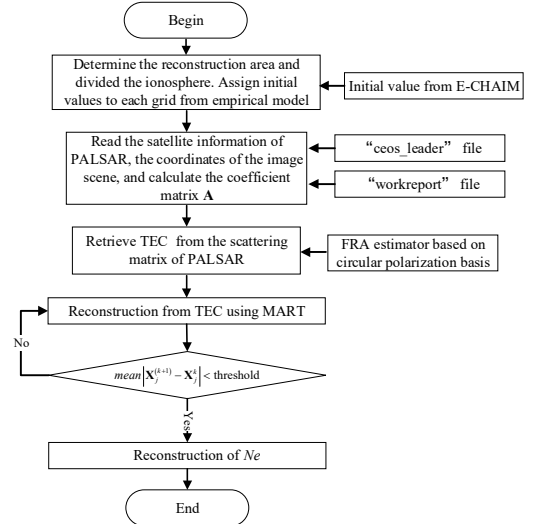
Equation (6) can be rewritten in matrix form, i.e.,

$$\mathbf{y}_{M \times 1} = \mathbf{A}_{M \times J} \cdot \mathbf{X}_{J \times 1} + \boldsymbol{\varepsilon}_{M \times 1}, \quad (7)$$

where  $\mathbf{y}$  is a column vector of the  $M$  corresponding to  $\text{TEC}_i$ ,



**Fig. 1.** Schematic map of CIT based on  $M$  PALSAR images. The ionosphere is divided into  $J$  voxels along the longitude, latitude, and altitude directions.



**Fig. 2.** The flowchart of the CIT based on PALSAR.

$\mathbf{A}$  is an  $M \times J$  matrix corresponding to  $a_{ij}$ ,  $\mathbf{X}$  denotes a column vector of  $J$  corresponding to  $x_j$ , and  $\boldsymbol{\varepsilon}$  is the error vector. Compared to the number of unknowns IED to be solved, the number of TEC values is inevitably insufficient, that is,  $M < J$ . This leads to an ill-posed problem in electron density reconstruction based on CIT. In order to solve  $\mathbf{X}$ , the multiplicative algebraic reconstruction technique (MART) is commonly used for CIT inversion, which can be written as

$$X_j^{(k+1)} = X_j^k \left( y_i / \left( \mathbf{X}^k \cdot \mathbf{a}_i \right) \right)^{\lambda_k a_{ij}}, \quad (8)$$

where  $X_j^{(k+1)}$  and  $X_j^k$  are the IED of the  $j$ th voxel at iteration  $k+1$  and  $k$ , respectively.  $\mathbf{a}_i$  is the  $i$ th row of matrix  $\mathbf{A}$ . The initial values of IED,  $\mathbf{X}^{(0)}$ , are given by the Empirical Canadian High Arctic Ionospheric Model (E-CHAIM) model [21].  $\lambda_k = 0.8$  is the relaxation factor.  $\langle \bullet \rangle$  and  $\|\bullet\|$  denotes the inner product and norm, respectively. The corresponding flowchart of the CIT proposed in this letter is shown in Fig. 2.

**Table 1.** Information of the PALSAR names, observation times, image center coordinates, corresponding PALSAR satellite coordinates for each image, and retrieved TEC.

| Group  | Data Name       | Observation Time<br>(UT:Y/M/D/HH/MM) | Image Center<br>Latitude (°) | Image Center<br>Longitude (°) | Satellite<br>Latitude (°) | Satellite<br>Longitude (°) | Satellite<br>Altitude (km) | TEC<br>(TECU) |
|--------|-----------------|--------------------------------------|------------------------------|-------------------------------|---------------------------|----------------------------|----------------------------|---------------|
| Group1 | ALPSRP241532220 | 2010/08/06/21/05                     | 68.542                       | -144.928                      | 67.543                    | -138.621                   | 693.872                    | 7.798         |
|        | ALPSRP241532230 | 2010/08/06/21/05                     | 68.066                       | -145.312                      | 67.089                    | -139.120                   | 693.913                    | 7.792         |
|        | ALPSRP241532240 | 2010/08/06/21/05                     | 67.584                       | -145.705                      | 66.635                    | -139.601                   | 693.955                    | 7.712         |
|        | ALPSRP241532250 | 2010/08/06/21/05                     | 67.108                       | -146.079                      | 66.179                    | -140.066                   | 693.999                    | 7.722         |
|        | ALPSRP241532260 | 2010/08/06/21/06                     | 66.629                       | -146.442                      | 65.723                    | -140.516                   | 694.043                    | 7.729         |
|        | ALPSRP241532270 | 2010/08/06/21/06                     | 66.149                       | -146.777                      | 65.208                    | -141.005                   | 694.094                    | 7.814         |
|        | ALPSRP241532280 | 2010/08/06/21/06                     | 65.669                       | -147.117                      | 64.749                    | -141.425                   | 694.140                    | 7.795         |
|        | ALPSRP241532290 | 2010/08/06/21/06                     | 65.183                       | -147.450                      | 64.289                    | -141.832                   | 694.187                    | 7.653         |
|        | ALPSRP241532300 | 2010/08/06/21/06                     | 64.711                       | -147.763                      | 63.828                    | -142.227                   | 694.235                    | 7.633         |
|        | ALPSRP241532310 | 2010/08/06/21/06                     | 64.222                       | -148.054                      | 63.366                    | -142.609                   | 694.283                    | 7.861         |
| Group2 | ALPSRP241532320 | 2010/08/06/21/06                     | 63.737                       | -148.357                      | 62.849                    | -143.028                   | 694.339                    | 7.949         |
|        | ALPSRP241532330 | 2010/08/06/21/06                     | 63.253                       | -148.650                      | 62.382                    | -143.388                   | 694.389                    | 7.978         |
|        | ALPSRP241532340 | 2010/08/06/21/07                     | 62.776                       | -148.917                      | 61.918                    | -143.738                   | 694.440                    | 8.072         |
|        | ALPSRP274271250 | 2011/03/19/07/32                     | 62.294                       | -146.679                      | 61.409                    | -151.744                   | 694.581                    | 1.465         |
|        | ALPSRP274271260 | 2011/03/19/07/32                     | 62.780                       | -146.957                      | 61.875                    | -152.082                   | 694.527                    | 1.539         |
|        | ALPSRP274271270 | 2011/03/19/07/32                     | 63.266                       | -147.243                      | 62.339                    | -152.430                   | 694.474                    | 1.520         |
|        | ALPSRP274271280 | 2011/03/19/07/32                     | 63.748                       | -147.512                      | 62.803                    | -152.788                   | 694.421                    | 1.489         |
|        | ALPSRP274271290 | 2011/03/19/07/32                     | 64.228                       | -147.810                      | 63.266                    | -153.157                   | 694.370                    | 1.671         |
|        | ALPSRP274271300 | 2011/03/19/07/32                     | 64.711                       | -148.120                      | 63.785                    | -153.584                   | 694.313                    | 1.763         |
|        | ALPSRP274271310 | 2011/03/19/07/32                     | 65.193                       | -148.439                      | 64.247                    | -153.977                   | 694.263                    | 1.704         |
| Group3 | ALPSRP274271320 | 2011/03/19/07/32                     | 65.674                       | -148.751                      | 64.707                    | -154.381                   | 694.214                    | 1.667         |
|        | ALPSRP274271330 | 2011/03/19/07/33                     | 66.153                       | -149.089                      | 65.166                    | -154.798                   | 694.166                    | 1.562         |
|        | ALPSRP274271340 | 2011/03/19/07/33                     | 66.632                       | -149.439                      | 65.624                    | -155.229                   | 694.119                    | 1.424         |
|        | ALPSRP274271350 | 2011/03/19/07/33                     | 67.119                       | -149.807                      | 66.139                    | -155.731                   | 694.067                    | 1.399         |
|        | ALPSRP274271360 | 2011/03/19/07/33                     | 67.596                       | -150.163                      | 66.594                    | -156.193                   | 694.021                    | 1.193         |
|        | ALPSRP274271370 | 2011/03/19/07/33                     | 68.065                       | -150.544                      | 67.049                    | -156.671                   | 693.977                    | 1.089         |
|        | ALPSRP276021270 | 2011/03/31/07/27                     | 63.256                       | -146.166                      | 62.330                    | -151.355                   | 694.436                    | 5.323         |
|        | ALPSRP276021280 | 2011/03/31/07/27                     | 63.745                       | -146.462                      | 62.793                    | -151.712                   | 694.384                    | 5.267         |
|        | ALPSRP276021290 | 2011/03/31/07/28                     | 64.230                       | -146.740                      | 63.314                    | -152.127                   | 694.326                    | 5.226         |
|        | ALPSRP276021300 | 2011/03/31/07/28                     | 64.713                       | -147.050                      | 63.776                    | -152.508                   | 694.275                    | 5.195         |
| Group3 | ALPSRP276021310 | 2011/03/31/07/28                     | 65.194                       | -147.369                      | 64.237                    | -152.900                   | 694.225                    | 5.228         |
|        | ALPSRP276021320 | 2011/03/31/07/28                     | 65.668                       | -147.692                      | 64.698                    | -153.304                   | 694.176                    | 5.426         |
|        | ALPSRP276021330 | 2011/03/31/07/28                     | 66.158                       | -148.021                      | 65.157                    | -153.721                   | 694.128                    | 5.749         |
|        | ALPSRP276021340 | 2011/03/31/07/28                     | 66.632                       | -148.368                      | 65.673                    | -154.206                   | 694.074                    | 5.828         |
|        | ALPSRP276021350 | 2011/03/31/07/28                     | 67.117                       | -148.734                      | 66.130                    | -154.653                   | 694.028                    | 5.938         |
|        | ALPSRP276021360 | 2011/03/31/07/28                     | 67.594                       | -149.089                      | 66.586                    | -155.114                   | 693.982                    | 6.232         |
|        | ALPSRP276021370 | 2011/03/31/07/29                     | 68.075                       | -149.480                      | 67.040                    | -155.592                   | 693.938                    | 6.443         |
|        | ALPSRP276021380 | 2011/03/31/07/29                     | 68.532                       | -149.865                      | 67.494                    | -156.087                   | 693.894                    | 6.619         |
|        | ALPSRP276021390 | 2011/03/31/07/29                     | 69.024                       | -150.296                      | 68.002                    | -156.665                   | 693.846                    | 6.407         |

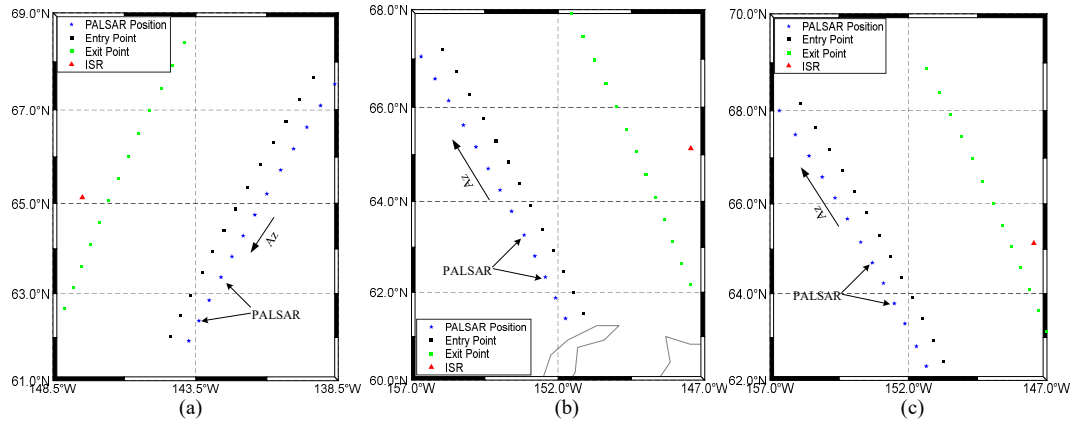
### III. APPLICATIONS OF CIT BASED ON PALSAR

In this section, three groups of PALSAR data from different times and locations in Alaska are used to validate its CIT capability [22]. Each group consists of 13 radar images. The detailed information for the three sets of images are shown in Table I. The data names, observation times, image center coordinates, satellite coordinates for each image, and TEC retrieval values are calculated. All the radar observations are acquired at a nadir angle of  $21.5^\circ$ . The observation duration of each group is no more than 3 minutes and it is reasonable not to account for ionospheric variation. The reason for selecting Alaska is that the  $\cos\theta_H$  in (2) approaches zero for PALSAR near the magnetic equator, the FRA can be ignored. However, in high-latitude regions like Alaska, where  $\cos\theta_H$  is close to 1, even with a low level of ionospheric activity, a significant FRA can still occur, allowing for reliable retrieval of TEC.

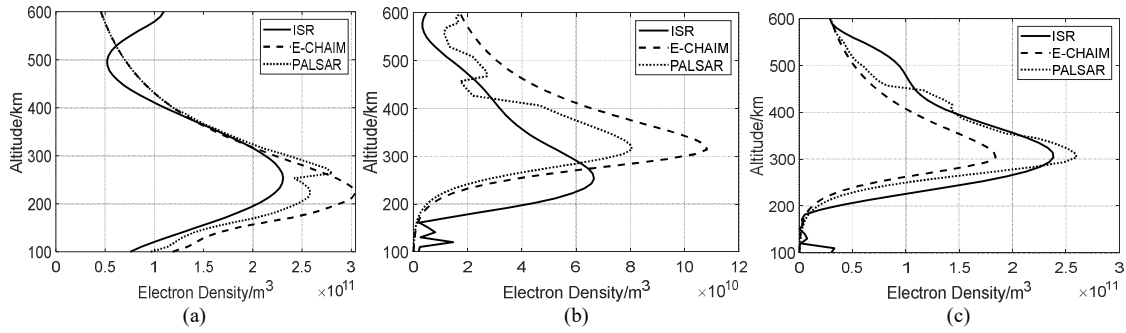
For the first group, the CIT range is set from  $138.5^\circ\text{W}$  to

$148.5^\circ\text{W}$  longitude, from  $61^\circ\text{N}$  to  $69^\circ\text{N}$  latitude, and from 100 to 600 km altitude. The initial IED for each voxel is obtained from the E-CHAIM at the same time as the PALSAR observation time, which is 21:06 UT on August 6, 2010. Fig. 3(a) illustrates the plan view of the CIT region, with PALSAR traversing in the azimuth direction and obtaining 13 images. The red triangle represents the position of the ISR ( $147.47^\circ\text{W}$ ,  $65.13^\circ\text{N}$ ), which is used as an independent and true value for comparison [23]. The voxel size is set to  $2.5^\circ \times 2^\circ \times 10\text{ km}$ . The blue pentagrams are the positions of 13 PALSAR images. The black and green squares mark the intersections of the PALSAR rays with the ionospheric layer at 600 km altitude (entry points) and 100 km altitude (exit points), respectively.

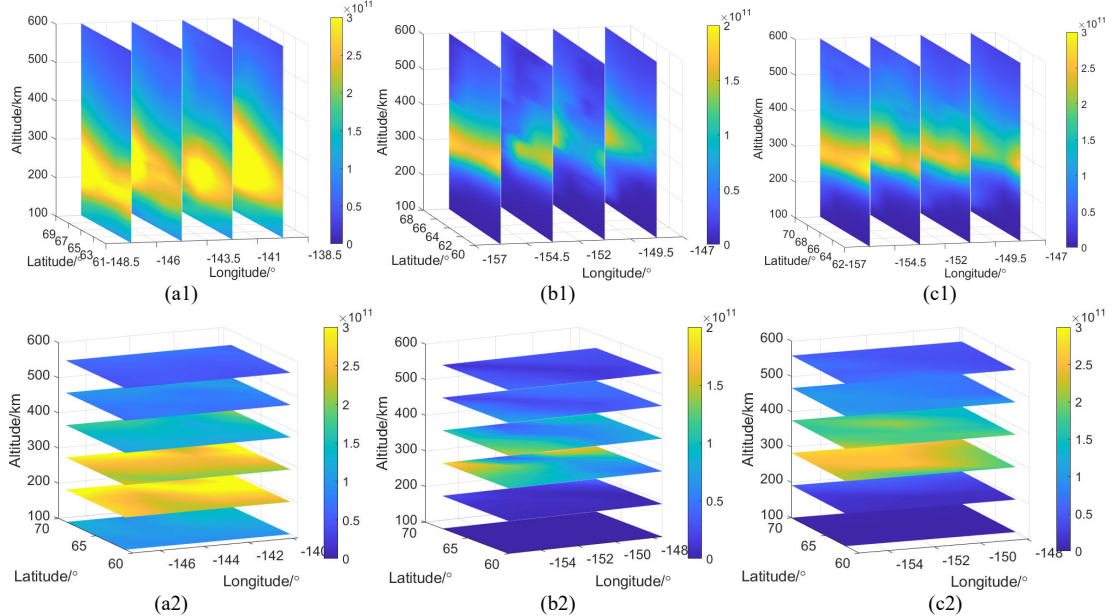
The ionospheric puncture point (IPP) is set as 300 km. The CIT is therefore made according to Section II and Fig. 2. The reconstruction profile in the altitude direction where the ISR located is illustrated in Fig. 4(a), which also includes the results of the ISR observation (21:08 UT on August 6, 2010) and the



**Fig. 3.** Three groups of the geographic relationships of the satellite positions (blue pentagram), intersections of PALSAR rays entering (black squares) and exiting (green squares) the ionosphere, the ISR (red triangle) and the plan view of the CIT region. The reconstruction time are 21:06 UT on August 6, 2010 (a), 07:30 UT on March 19, 2011 (b), and 07:28 UT on March 31, 2011 (c).



**Fig.4.** The reconstruction profiles based on three sets of PALSAR measured data (thin dashed lines) and E-CHAIM (thick dashed lines) at the ISR position. The black solid line represents the ISR observation, which represents the true value. The reconstruction time are 21:06 UT on August 6, 2010 (a), 07:30 UT on March 19, 2011 (b), and 07:28 UT on March 31, 2011 (c).



**Fig.5.** 3-D reconstructions based on the three groups of PALSAR data. (a1), (b1) and (c1) respectively show the variations of the three reconstructions with longitude. (a2), (b2), and (c2) respectively show the variations of the three reconstructions with altitude.

E-CHAIM model. The average deviation (AD) is used for evaluation, which is defined as

$$AD = \text{mean} |x_{j,ISR} - x_j|, \quad (9)$$

where  $x_{j,ISR}$  is the IED profile from the ISR, and  $x_j$  is obtained from the PALSAR or E-CHAIM. It can be seen that after the CIT using PALSAR, the IED is closer to the true value, with an AD of  $2.29 \times 10^{10} \text{ el/m}^3$ , while the AD of the E-CHAIM is  $3.22 \times 10^{10} \text{ el/m}^3$ , which is an increase of 28.9%.

For the second group, the reconstruction area is set from  $147^\circ\text{W}$  to  $157^\circ\text{W}$  longitude, from  $60^\circ\text{N}$  to  $68^\circ\text{N}$  latitude, and from 100 to 600 km altitude. Other settings are the same as the first group, and its plan view is shown in Fig. 3(b). The ISR observation time is 07:30 UT on March 19, 2011. The result is shown in Fig. 4(b), with an AD of  $1.69 \times 10^{10} \text{ el/m}^3$ , while the AD of the E-CHAIM is  $2.53 \times 10^{10} \text{ el/m}^3$ , which is an increase of 33.2%. It can be seen that although the reconstructed has been significantly improved compared with the E-CHAIM, there is still obvious difference from the true value. The reason is that CIT is highly dependent on the initial value. If the initial value has a large deviation, even if the TEC values retrieved by the PALSAR are accurate, the results will still be biased.

As shown in Fig.3(c), the reconstruction area of the third group is set from  $147^\circ\text{W}$  to  $157^\circ\text{W}$  longitude, from  $62^\circ\text{N}$  to  $70^\circ\text{N}$  latitude, from 100 to 600 km altitude, and other settings remain unchanged. The ISR observation time is 07:28 UT on March 31, 2011. The reconstruction result is shown in Fig. 4(c), with an AD of  $1.95 \times 10^{10} \text{ el/m}^3$ . In comparison, the AD of the E-CHAIM model is  $3.72 \times 10^{10} \text{ el/m}^3$ , which represents a increase of 47.6%. It can be seen that the PALSAR-based results in Fig.4 are non-monotonic and non-smooth. This is due to the limited number and elevation angle of PALSAR rays, which means that not all IEDs corresponding to a specific height are adequately modulated. However, this does not affect the overall results being better than the initial value. Fig.5 presents the 3-D results of IED along the longitude and altitude. It can be seen that for different cross-sections, there is a high IED region between 200 km and 400 km, which corresponds to the results shown in Fig. 4, and the IED shows an increasing trend followed by a decrease as the altitude increases.

#### IV. SUMMARY AND CONCLUSIONS

When the PALSAR operates for Earth observation, it is influenced by both the ionosphere and the geomagnetic field, especially in high-latitude regions. On the other hand, this also provides a new means for polar ionospheric sounding. Based on it, this letter investigates a study of CIT using fully polarimetric PALSAR. Experimental validations show that the IED reconstructed by the PALSAR is closer to the true value compared to empirical models. This demonstrates that PALSAR not only has the capability to detect horizontal structure (i.e., FRA or TEC) but also has the ability to reconstruct vertical electron density. In addition, compared with GPS, PALSAR signals propagate in two-way, and do not require ground receiving station, so PALSAR method is easier for implementation. The detection of polar ionospheric anomaly, such as auroral arc, is meaningful for understanding the polar ionosphere. Our research will focus more on such phenomena in the future.

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